



KINROSS WOLAROI  
SCHOOL

**2025**

## **Year 12 Mathematics Extension 2**

Trial HSC Examination

Teacher Setting Paper: Mr Bowman

Head of Department: Mr Doyle

### **General Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculator may be used
- Write your answers for Section I on the multiple-choice answer sheet provided
- Write your answers for Section II in the booklets provided. Extra writing booklets are available.
- A reference sheet is provided with this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations.

**Total marks – 100**

### **Section I – Multiple-Choice**

10 marks

Attempt Questions 1-10

Allow 15 minutes for this section

### **Section II – Free Response**

90 marks

Attempt questions 11 – 16

Allow 2 hours and 45 minutes for this section

*This examination paper does not necessarily reflect the content or format of the Higher School Certificate Examination in this subject.*

**Section I****10 marks****Attempt Questions 1 – 10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Questions 1 - 10.**QUESTION 1** $\omega$  is a complex fifth root of unity. What is the value of  $\omega^{2025}$ ?

- (A) 1
- (B)  $-1$
- (C)  $i$
- (D)  $-i$

**QUESTION 2**

What is the contrapositive of this statement?

“If it is snowing, then the ground is white.”

- (A) If the ground is white, then it is snowing.
- (B) If the ground is not white, then it is not snowing.
- (C) If the ground is not white, then it is snowing.
- (D) If it is not snowing, then the ground is not white.

**QUESTION 3**Which equation describes the line joining  $A(4, 6, 5)$  to  $B(-2, 3, -1)$ ?

- (A)  $r = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} - \lambda \begin{pmatrix} -2 \\ 3 \\ -5 \end{pmatrix}$
- (B)  $r = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} - \lambda \begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix}$
- (C)  $r = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -3 \\ -6 \end{pmatrix}$
- (D)  $r = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix}$

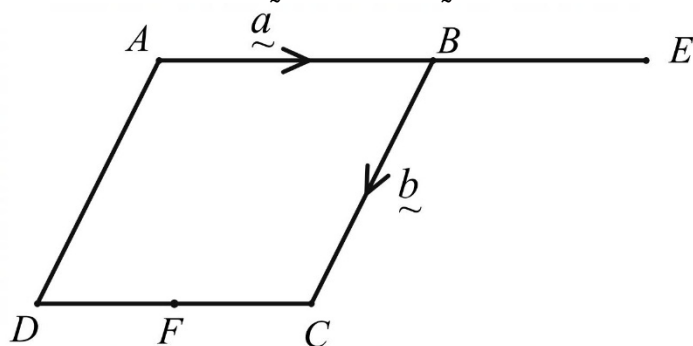
**QUESTION 4**

A particle is describing Simple Harmonic Motion in a straight line with an amplitude of 3 metres. When the particle is 2 metres from the centre of the motion, its speed is 5 m/s. What is the period of the motion?

- (A)  $\frac{\sqrt{5}\pi}{2}$
- (B)  $\frac{2\sqrt{5}\pi}{5}$
- (C)  $\sqrt{5}\pi$
- (D)  $2\sqrt{5}\pi$

**QUESTION 5**

$ABCD$  is a rhombus.  $\overrightarrow{AB} = \underline{a}$  and  $\overrightarrow{BC} = \underline{b}$ .



$\overrightarrow{BE}$  is an extension of  $\overrightarrow{AB}$ , such that  $\overline{AB} : \overline{BE} = 4 : 3$ .  
The point  $F$  is the midpoint of  $CD$ .

Which expression gives the vector  $\overrightarrow{FE}$  in terms of  $\underline{a}$  and  $\underline{b}$ ?

- (A)  $\frac{5}{4} \underline{a} - \underline{b}$
- (B)  $\frac{5}{4} \underline{a} + \underline{b}$
- (C)  $\frac{11}{6} \underline{a} - \underline{b}$
- (D)  $\frac{11}{6} \underline{a} + \underline{b}$

**QUESTION 6**

A cubic polynomial  $P(z)$  has all real coefficients.

$z = 3$  and  $z = 2 + i$  are two roots of  $P(z) = 0$ .

Which equation describes  $P(z)$ ?

- (A)  $P(z) = z^3 - 7z^2 - 17z + 15$
- (B)  $P(z) = z^3 - 7z^2 + 17z - 15$
- (C)  $P(z) = z^3 + 7z^2 - 35z + 15$
- (D)  $P(z) = z^3 + 7z^2 + 35z - 15$

**QUESTION 7**

Issy tries to find  $\int_{-2}^0 \frac{1}{x^2 - 1} dx$  but she can't seem to get a simple answer.

Which of the following could be the reason she is having trouble?

- (A) Because the function is not differentiable at  $x = 0$ .
- (B) Because there is no appropriate substitution to begin calculating the integral.
- (C) Because the function has a horizontal asymptote in the domain  $[-2, 0]$ .
- (D) Because the function is discontinuous in the domain  $[-2, 0]$ .

**QUESTION 8**

Ollie is trying to find  $\int \sqrt{4x^2 - 9} dx$ . Which substitution should he use?

- (A)  $x = \sec \theta$
- (B)  $x = \frac{3}{2} \sec \theta$
- (C)  $x = \frac{2}{3} \sec \theta$
- (D)  $x = \frac{3}{2} \sin \theta$

**QUESTION 9**

If  $a$ ,  $b$ ,  $c$  and  $d$  are consecutive integers, then which of the following statements may be false ?

- (A)  $a + b + c + d$  is divisible by 2
- (B)  $a + b + c + d$  is divisible by 4
- (C)  $a \times b \times c \times d$  is divisible by 3
- (D)  $a \times b \times c \times d$  is divisible by 8

**QUESTION 10**

A particle starts moving along a straight line from rest at the origin. At time  $t$ , its velocity is  $v$  and its displacement from the origin is  $x$ .

If  $\frac{dv}{dx} = \frac{1}{2v}$ , which of the following best describes the particle's acceleration and velocity?

- (A) Constant acceleration and constant velocity
- (B) Constant acceleration and decreasing velocity
- (C) Constant acceleration and increasing velocity
- (D) Increasing acceleration and increasing velocity

**END OF SECTION I**

**Section II****90 marks****Attempt Questions 11 – 16****Allow about 2 hours 45 minutes for this section**

Answer each question in the booklets provided. Extra writing booklets are available.

|                                                                                                                                                    | <b>Marks</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>QUESTION 11</b> (14 marks) <b>Start a new writing booklet</b>                                                                                   |              |
| (a) (i) Write $1 - i\sqrt{3}$ in mod-arg form.                                                                                                     | <b>1</b>     |
| (ii) Hence write $\frac{(1-i\sqrt{3})^4}{1+i\sqrt{3}}$ in exact Cartesian form.                                                                    | <b>3</b>     |
| (b) Find a vector equation for the line that passes through $A(1, -1, 0)$ and $B(2, -1, 4)$ .                                                      | <b>2</b>     |
| (c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x} dx$ .                                                                                    | <b>3</b>     |
| (d) Prove that $\sqrt{2}$ is irrational by contradiction.                                                                                          | <b>3</b>     |
| (e) Let $\underline{u} = 5\underline{i} + \underline{j} + 3\underline{k}$ and $\underline{v} = -2\underline{i} - 2\underline{j} + \underline{k}$ . |              |
| (i) Find the dot product $\underline{u} \cdot \underline{v}$                                                                                       | <b>1</b>     |
| (ii) Find the unit vector $\hat{v}$ .                                                                                                              | <b>1</b>     |

**QUESTION 12** (16 marks) **Start a new writing booklet**

(a) (i) Show that  $(1 - 3i)^2 = -8 - 6i$ . 1

(ii) Hence, or otherwise, solve the equation  $2z^2 - 8z + (12 + 3i) = 0$ . 2

(b) The points  $D, E$  and  $F$  have position vectors relative to the origin  $O$  given by 4

$$\overrightarrow{OD} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix}, \overrightarrow{OE} = \begin{pmatrix} 6 \\ 3 \\ -1 \end{pmatrix}, \overrightarrow{OF} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}.$$

Find  $\angle DEF$  to the nearest minute.

(c) Sketch, on an Argand Diagram, the region that satisfies 2

$$|z - 1| \leq 3.$$

(d) A light string passes over a smooth pulley. 3

Attached to one end is a 9 kg mass, and to the other end, a 7 kg mass.

Find the acceleration of the 9 kg mass in terms of  $g$ .

(e) (i) Express  $\frac{x^3 + 3x^2 - 11x + 12}{(x-2)^2(x^2+1)}$  in the form  $\frac{a}{(x-2)^2} + \frac{b}{x-2} + \frac{cx+d}{x^2+1}$ . 2

(ii) Hence evaluate 2

$$\int_0^1 \frac{x^3 + 3x^2 - 11x + 12}{(x-2)^2(x^2+1)} dx.$$

Leave your answer in exact form.

**QUESTION 13 (14 marks) Start a new writing booklet**

(a) A particle travels in a straight line so that its velocity  $v$  cm/s and displacement  $x$  cm are related by the equation  $v = -0.2x$ .

(i) Determine the acceleration  $a$  in terms of its displacement  $x$ . **2**

(ii) Is the particle moving in simple harmonic motion? **1**  
Justify your answer.

(iii) It is known that the initial displacement of the particle is  $x = 4$  cm.  
Determine, correct to 2 decimal places, when the particle  
has a displacement of 2 cm. **2**

(b) The area enclosed by the graph of  $y = \sqrt{x} \sin x$ , the  $x$ -axis and  
the lines  $x = 0, x = \pi$  is rotated around the  $x$ -axis. **4**

Find the exact volume of the solid formed.

(c) Consider the polynomial  $P(z) = 2z^5 - 13z^4 + 52z^3 - 91z^2 + 86z - 26$ . **5**

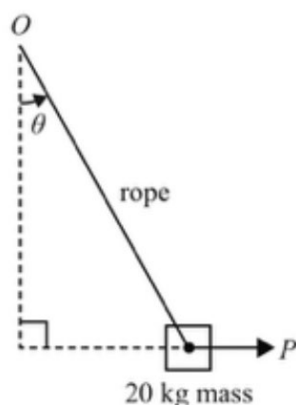
Given  $z = 1 - i$  is a zero of  $P(z)$ , find all the roots of  $P(z) = 0$ .

**QUESTION 14 (15 marks) Start a new writing booklet**

- (a) A taut (tightly pulled) rope of length  $1\frac{2}{3}$  m suspends a mass of 20 kg from a fixed point  $O$ .

A horizontal force of  $P$  newtons displaces the mass by 1 metre horizontally so that the taut rope is then at an angle  $\theta$  to the vertical.

- (i) Copy the diagram below into your writing booklet and show clearly all the forces acting on the mass. 1



- (ii) Show that  $\sin\theta = \frac{3}{5}$ . 1

- (iii) Find the magnitude of the tension force in the rope in newtons (Leave your answer in terms of  $g$ ). 3

- (b) Prove that, if  $a^2 - 2a + 2025$  is even, then  $a$  is odd. 3

- (c) Use integration by parts to find an expression for 3
- $$\int e^x \sin x \, dx.$$

- (d) Use the  $t$ -formulas to find an expression for 4
- $$\int \frac{1}{5 + 4 \cos x + 3 \sin x} \, dx.$$

**QUESTION 15** (16 marks) **Start a new writing booklet**

- (a)  $OABC$  is a kite, with  $OB$  as its line of symmetry.  $|OA| = |OC|$ .  $|AB| = |BC|$ .

Let  $\underline{a} = \overrightarrow{OA}$ ,  $\underline{b} = \overrightarrow{OB}$ ,  $\underline{c} = \overrightarrow{OC}$ .

- (i) Prove that  $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c}$ . 1

- (ii) Hence or otherwise, prove that the diagonals of a kite are perpendicular. 3

- (b) Use the principle of mathematical induction to prove that 3

$$(1 + x)^n > 1 + nx \text{ for } n > 1 \text{ and } x > 0$$

- (c) Kajan decides to go bungy-jumping. This involves being tied to a bridge at point  $O$  by an elastic cable of length  $L$  metres, and then falling vertically from rest from this point.

After Kajan free falls  $L$  metres, he is slowed down by the cable, which exerts a force (in newtons) of  $mgk$  times the distance greater than  $L$  that he has fallen (where  $m$  is his mass in kg and  $k$  is a constant).

Let  $x$  metres be the distance Kajan has fallen, and let  $v$  m/s be his speed at  $x$ .

- (i) Show that  $\ddot{x} = \begin{cases} g & x \leq L \\ g - gk(x - L) & x > L \end{cases}$  2

- (ii) Show that  $v^2 = 2gL$  at the instant Kajan is at  $x = L$ . 1

- (iii) Show that  $v^2 = 2gx - kg(x - L)^2$  for  $x > L$ . 1

- (iv) Show that Kajan stops for the first time at  $x = L + \frac{1}{k} + \sqrt{\frac{2L}{k} + \left(\frac{1}{k}\right)^2}$ . 3

- (v) Given that  $\frac{1}{k} = \frac{L}{4}$ , show that  $O$  must be at least  $2L$  metres above any 2

obstruction on Kajan's path.

**QUESTION 16 (15 marks) Start a new writing booklet**

(a) Evaluate

**4**

$$\int_1^2 \frac{x+1}{\sqrt{3x-2-x^2}} dx$$

(b) Find  $|\underline{u} + \underline{v}|$  if  $|\underline{u}| = 6$ ,  $|\underline{v}| = 5$  and  $\underline{u} \cdot \underline{v} = -4$ .**3**

(c) Prove that

**2**

$$\frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} = \frac{(n+1)!}{r!(n-r+1)!}.$$

(d) It is known that the arithmetic mean of three numbers is greater than the geometric mean of those numbers, i.e.

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}.$$

(i) Given  $a+b+c=t$  and  $a, b, c > 0$ , prove that**3**

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{t}.$$

(ii) Hence, given  $a, b, c > 0$  and  $a+b+c=1$ , prove that**3**

$$\left(\frac{1}{a} - 1\right) \left(\frac{1}{b} - 1\right) \left(\frac{1}{c} - 1\right) \geq 8.$$

**END OF PAPER**

Student Name: \_\_\_\_\_

**2025****Year 12 Mathematics Extension 2**

Trial HSC Examination

**MULTIPLE-CHOICE ANSWER SHEET**

For multiple choice questions, choose the best answer A, B, C or D and fill in the correct circle.

1.    ☐ A   ☐ B   ☐ C   ☐ D
2.    ☐ A   ☐ B   ☐ C   ☐ D
3.    ☐ A   ☐ B   ☐ C   ☐ D
4.    ☐ A   ☐ B   ☐ C   ☐ D
5.    ☐ A   ☐ B   ☐ C   ☐ D
6.    ☐ A   ☐ B   ☐ C   ☐ D
7.    ☐ A   ☐ B   ☐ C   ☐ D
8.    ☐ A   ☐ B   ☐ C   ☐ D
9.    ☐ A   ☐ B   ☐ C   ☐ D
10.   ☐ A   ☐ B   ☐ C   ☐ D

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# SOLUTIONS

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**QUESTION 2**

What is the contrapositive of this statement?

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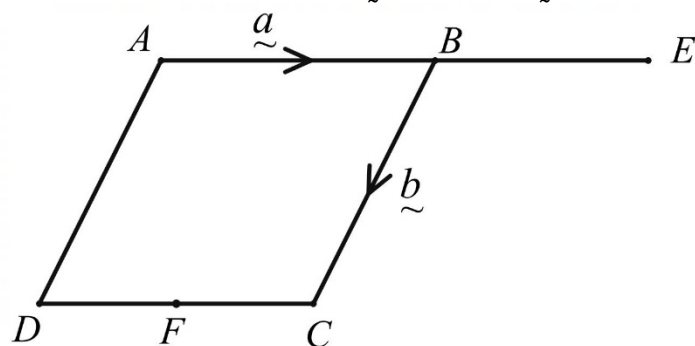
(B)  $\frac{2\sqrt{5}\pi}{5}$

(C)  $\sqrt{5}\pi$

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Issy tries to find  $\int_{-2}^0 \frac{1}{x^2 - 1} dx$  but she can't seem to get a simple answer.

Which of the following could be the reason she is having trouble?

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If  $a, b, c$  and  $d$  are consecutive integers, then which of the following statements may be false ?

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- (D)  $a \times b \times c \times d$  is divisible by 8

**QUESTION 10**

A particle starts moving along a straight line from rest at the origin. At time  $t$ , its velocity is  $v$  and its displacement from the origin is  $x$ .

If  $\frac{dv}{dx} = \frac{1}{2v}$ , which of the following best describes the particle's acceleration and velocity?

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**END OF SECTION I**

**Section II****90 marks****Attempt Questions 11 – 16****Allow about 2 hours 45 minutes for this section**

Answer each question in the booklets provided. Extra writing booklets are available.

**Marks****QUESTION 11 (15 marks) Start a new writing booklet**

- (a) (i) Write
- $1 - i\sqrt{3}$
- in mod-arg form.

**1**

$$\begin{aligned}
 1 - i\sqrt{3} &= 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) \\
 &= 2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right) \\
 &= 2\operatorname{cis}\left(-\frac{\pi}{3}\right)
 \end{aligned}$$

- (ii) Hence write
- $\frac{(1-i\sqrt{3})^4}{1+i\sqrt{3}}$
- in exact Cartesian form.

**3**

$$\begin{aligned}
 \frac{(1 - i\sqrt{3})^4}{1 + i\sqrt{3}} &= \frac{\left(2\operatorname{cis}\left(-\frac{\pi}{3}\right)\right)^4}{2\operatorname{cis}\frac{\pi}{3}} \\
 &= \frac{16\operatorname{cis}\left(-\frac{4\pi}{3}\right)}{2\operatorname{cis}\left(\frac{\pi}{3}\right)} \\
 &= \frac{16\operatorname{cis}\left(\frac{2\pi}{3}\right)}{2\operatorname{cis}\left(\frac{\pi}{3}\right)} \\
 &= 8\operatorname{cis}\left(\frac{\pi}{3}\right) \\
 &= 8\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) \\
 &= 4 + 4i\sqrt{3}
 \end{aligned}$$

- (b) Find the vector equation of a line that passes through
- $A(1, -1, 0)$
- and
- $B(2, -1, 4)$
- .

**2**

$$r = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$$

(c) Evaluate

3

$$\begin{aligned}
 & \int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x} dx. \\
 &= \int_0^1 \frac{2dt}{1 + \frac{1-t^2}{1+t^2}} \\
 &= \int_0^1 \frac{2dt}{1+t^2+1-t^2} \\
 &= \int_0^1 1 dt \\
 &= [t]_0^1 \\
 &= 1
 \end{aligned}$$

(d) Prove that  $\sqrt{2}$  is irrational by contradiction.

3

Assume rational, so

$$\begin{aligned}
 \sqrt{2} &= \frac{p}{q} \quad (p, q \text{ in lowest terms}) \\
 2 &= p^2/q^2 \\
 p^2 &= 2q^2 \\
 \therefore p^2 &\text{ is even} \\
 \therefore p &\text{ is even} \\
 p &= 2r \\
 p^2 &= 4r^2 \\
 2q^2 &= 4r^2 \\
 q^2 &= 2r^2 \\
 \therefore q^2 &\text{ is even} \\
 \therefore q &\text{ is even} \\
 \text{Both } p, q &\text{ are even} \\
 \therefore \frac{p}{q} &\text{ not in lowest terms.} \\
 \therefore \sqrt{2} &\text{ is irrational}
 \end{aligned}$$

(e) Let  $\mathbf{u} = 5\mathbf{i} + \mathbf{j} + 3\mathbf{k}$  and  $\mathbf{v} = -2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ .(i) Find the dot product  $\mathbf{u} \cdot \mathbf{v}$ 

1

$$\begin{aligned}
 \mathbf{u} \cdot \mathbf{v} &= 5(-2) + 1(-2) + 3(1) \\
 &= -9
 \end{aligned}$$

(ii) Find the unit vector  $\hat{\mathbf{v}}$ .

1

$$\begin{aligned}
 |\mathbf{v}| &= \sqrt{9} = 3 \\
 \therefore \hat{\mathbf{v}} &= -\frac{2}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} + \frac{1}{3}\mathbf{k}
 \end{aligned}$$

**QUESTION 12 (16 marks) Start a new writing booklet**

- (a) (i) Show that  $(1 - 3i)^2 = -8 - 6i$ . 1

$$\begin{aligned}(1 - 3i)^2 &= 1 - 6i - 9 \\ &= -8 - 6i \text{ as required}\end{aligned}$$

- (ii) Hence or otherwise solve the equation  $2z^2 - 8z + (12 + 3i) = 0$ . 2

$$\begin{aligned}z &= \frac{8 \pm \sqrt{64 - 8(12 + 3i)}}{4} \\ &= \frac{8 \pm \sqrt{-32 - 24i}}{4} \\ &= \frac{8 \pm 2\sqrt{-8 - 6i}}{4} \\ &= \frac{8 \pm 2(1 - 3i)}{4} \text{ using part (i)} \\ &= \frac{8 + 2 - 6i}{4} \text{ or } \frac{8 - 2 + 6i}{4} \\ &= \frac{5}{2} - \frac{3}{2}i \text{ or } \frac{3}{2} + \frac{3}{2}i\end{aligned}$$

- (b) The points  $D, E$  and  $F$  have position vectors relative to the origin  $O$  given by 4

$$\overrightarrow{OD} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix}, \overrightarrow{OE} = \begin{pmatrix} 6 \\ 3 \\ -1 \end{pmatrix}, \overrightarrow{OF} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}.$$

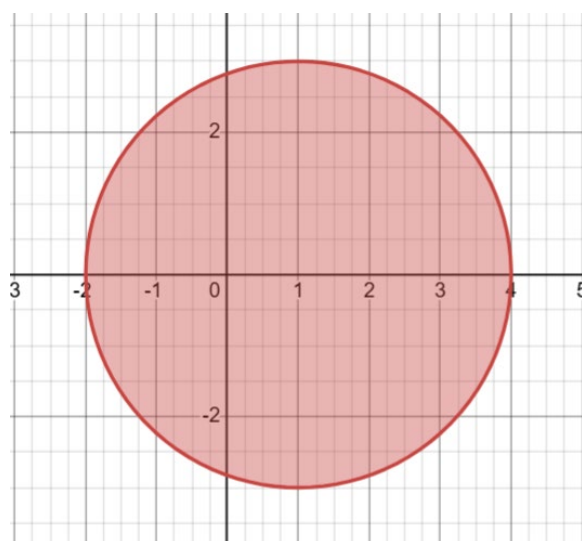
Find  $\angle DEF$  to the nearest minute.

$$\begin{aligned}\overrightarrow{ED} &= \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \\ \overrightarrow{EF} &= \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \\ |ED| &= 3 \\ |EF| &= \sqrt{6} \\ \cos\theta &= \frac{\overrightarrow{ED} \cdot \overrightarrow{EF}}{|\overrightarrow{ED}||\overrightarrow{EF}|} \\ &= -\frac{2}{3\sqrt{6}} \\ \theta &= 105^\circ 48'\end{aligned}$$

(c) Sketch, on an Argand Diagram, the region that satisfies

2

$$|z - 1| \leq 3.$$



(d) A light string passes over a smooth pulley.

3

Attached to one end is a 9 kg mass, and to the other end, a 7kg mass.

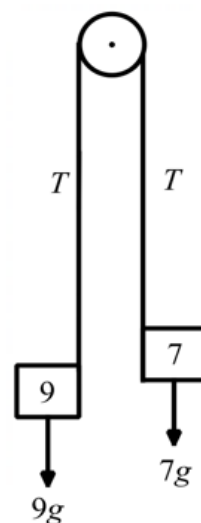
Find the acceleration of the 9 kg mass in terms of  $g$ .

$$9 \text{ kg mass: } 9\ddot{x} = 9g - T \dots (i)$$

$$7 \text{ kg mass: } 7\ddot{x} = T - 7g \dots (ii)$$

$$(i) + (ii) \rightarrow 16\ddot{x} = 2g$$

$$\ddot{x} = \frac{g}{8} \text{ (accepting } -\frac{g}{8})$$



(e) Evaluate

$$\int_0^1 \frac{x^3 + 3x^2 - 11x + 12}{(x-2)^2(x^2+1)} dx.$$

leaving your answer in exact form.

$$\text{Let } \frac{x^3 + 3x^2 - 11x + 12}{(x-2)^2(x^2+1)} = \frac{a}{(x-2)^2} + \frac{b}{x-2} + \frac{cx+d}{x^2+1}$$

$$x^3 + 3x^2 - 11x + 12 = a(x^2+1) + b(x-2)(x^2+1) + (cx+d)(x-2)^2$$

$$\text{Let } x = 2 \rightarrow 10 = 5a \rightarrow \mathbf{a = 2}$$

Equate coefficients of  $x^3$ 

$$1 = b + c \dots (i)$$

Equate coefficients of  $x^2$ 

$$3 = a - 2b + d - 4c$$

$$3 = 2 - 2b + d - 4c$$

$$d - 2b - 4c = 1 \dots (ii)$$

Equate constants

$$12 = a - 2b + 4d$$

$$4d - 2b = 10 \dots (iii)$$

From (i),

$$c = 1 - b$$

(ii):

$$d - 2b - 4(1 - b) = 1$$

$$2b + d = 5 \dots (iv)$$

(iii):

$$2d - b = 5$$

$$4d - 2b = 10 \dots (v)$$

(iv)+(v):

$$\mathbf{d = 3}$$

$$\mathbf{b = 1}$$

$$\mathbf{c = 0}$$

$$\therefore \frac{x^3 + 3x^2 - 11x + 12}{(x-2)^2(x^2+1)} = \frac{2}{(x-2)^2} + \frac{1}{x-2} + \frac{3}{x^2+1}$$

$$\begin{aligned} \int_0^1 \frac{x^3 + 3x^2 - 11x + 12}{(x-2)^2(x^2+1)} dx &= \int_0^1 \frac{2}{(x-2)^2} + \frac{1}{x-2} + \frac{3}{x^2+1} dx \\ &= \left[ \frac{-2}{x-2} + \ln|x-2| + 3 \tan^{-1} x \right]_0^1 \\ &= \left( \frac{-2}{-1} + 0 + \frac{3\pi}{4} \right) - \left( \frac{-2}{-2} + \ln 2 + 0 \right) \\ &= 1 + \frac{3\pi}{4} - \ln 2 \end{aligned}$$

**QUESTION 13 (14 marks) Start a new writing booklet**

- (a) A particle travels in a straight line so that its velocity  $v$  cm/s and displacement  $x$  cm are related by the equation  $v = -0.2x$ .

- (i) Determine the acceleration  $a$  in terms of its displacement  $x$ .

**2**

$$\begin{aligned}\ddot{x} &= \frac{d}{dx} \left( \frac{1}{2} v^2 \right) \\ \ddot{x} &= \frac{d}{dx} \left( \frac{1}{2} (-0.2x)^2 \right) = \frac{d}{dx} (0.04x^2) \\ \ddot{x} &= 0.08x\end{aligned}$$

- (ii) Is the particle moving in simple harmonic motion?  
Justify your answer.

**1**

NO because equation is not of format  $\ddot{x} = -n^2x$

- (iii) It is known that the initial displacement of the particle is  $x = 4$  cm. Determine, correct to 2 decimal places, when the particle has a displacement of 2 cm.

**2**

$$\begin{aligned}\frac{dx}{dt} &= -0.2x = -\frac{x}{5} \\ \frac{dx}{x} &= -\frac{1}{5} dt \\ t &= -5 \ln x + C \\ 0 &= -5 \ln 4 + C \\ C &= 5 \ln 4 \\ t &= 5 \ln \left( \frac{4}{x} \right) \\ \text{When } x &= 2, \\ t &= 5 \ln 2 = 3.47\end{aligned}$$

Particle has displacement of 2 cm after 3.47 sec.

- (b) The area enclosed by the graph of  $y = \sqrt{x} \sin x$ , the  $x$ -axis and the lines  $x = 0, x = \pi$  is rotated around the  $x$ -axis.

4

Find the exact volume of the solid formed.

$$V = \pi \int_0^{\pi} y^2 dx$$

$$= \pi \int_0^{\pi} x \sin^2 x dx$$

By parts:  $u = x, v' = \sin^2 x$   
 $u' = 1, v = \frac{x}{2} - \frac{\sin 2x}{4}$

$$V = \pi \left[ \frac{x^2}{2} - \frac{x \sin 2x}{4} \right]_0^{\pi} - \pi \int_0^{\pi} \frac{x}{2} - \frac{1}{4} \sin 2x dx$$

$$= \pi \left[ \frac{x^2}{2} - \frac{x \sin 2x}{4} \right]_0^{\pi} - \pi \left[ \frac{x^2}{4} + \frac{1}{8} \cos 2x \right]_0^{\pi}$$

$$= \pi \left( \frac{\pi^2}{2} - 0 - 0 + 0 \right) - \pi \left( \frac{\pi^2}{4} + \frac{1}{8} - 0 - \frac{1}{8} \right)$$

$$= \frac{\pi^3}{2} - \frac{\pi^3}{4}$$

$$= \frac{\pi^3}{4} \text{ cu units.}$$

(c) Consider the polynomial  $P(z) = 2z^5 - 13z^4 + 52z^3 - 91z^2 + 86z - 26$ .

**5**

Given  $z = 1 - i$  is a zero of  $P(z)$ , find all the roots of  $P(z) = 0$ .

$z = 1 - i$  is a zero, so  $z = 1 + i$  is a zero as well.

Hence the quadratic  $z^2 - 2z + 2$  is a factor of  $P(z)$ .

$$\text{Let } 2z^5 - 13z^4 + 52z^3 - 91z^2 + 86z - 26 = (z^2 - 2z + 2)(2z^3 + az^2 + bz - 13)$$

Equating coefficients of  $z^4$ :

$$a - 4 = -13 \rightarrow a = -9$$

Equating coefficients of  $z^3$ :

$$\begin{aligned} b - 2a + 4 &= 52 \\ b + 22 &= 52 \rightarrow b = 30 \end{aligned}$$

$$\therefore 2z^5 - 13z^4 + 52z^3 - 91z^2 + 86z - 26 = (z^2 - 2z + 2)(2z^3 - 9z^2 + 30z - 13)$$

Now for factors of  $2z^3 - 9z^2 + 30z - 13$ :

$$P\left(\frac{1}{2}\right) = 0, \text{ hence } 2z - 1 \text{ is a factor of } 2z^3 - 9z^2 + 30z - 13$$

$$\text{Let } 2z^3 - 9z^2 + 30z - 13 = (2z - 1)(z^2 + cz + 13)$$

Equating coefficients of  $z^2$ :

$$2c - 1 = -9 \rightarrow c = -4$$

$$\therefore 2z^3 - 9z^2 + 30z - 13 = (2z - 1)(z^2 - 4z + 13)$$

$$\text{The roots of } z^2 - 4z + 13 \text{ are } z = \frac{4 \pm \sqrt{-36}}{2} = z = 2 \pm 3i$$

So the 5 roots of  $P(z) = 0$  are:

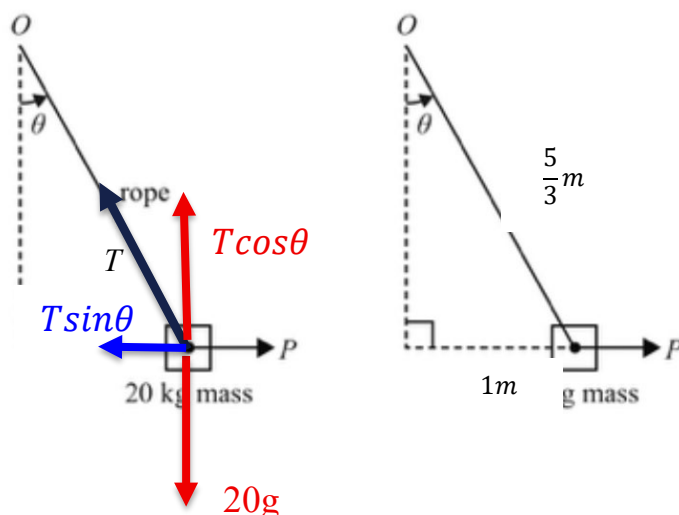
$$z = 1 - i, \quad 1 + i, \quad \frac{1}{2}, \quad 2 + 3i, \quad 2 - 3i$$

**QUESTION 14 (15 marks) Start a new writing booklet**

- (a) A taut (tightly pulled) rope of length  $1\frac{2}{3}$  m suspends a mass of 20 kg from a fixed point  $O$ .

A horizontal force of  $P$  newtons displaces the mass by 1 metre horizontally so that the taut rope is then at an angle  $\theta$  to the vertical.

- (i) Copy the diagram below into your writing booklet and show clearly all the forces acting on the mass. 1



- (ii) Show that  $\sin\theta = \frac{3}{5}$ . 1

From diagram at right,

$$\sin\theta = \frac{1}{\left(\frac{5}{3}\right)} = \frac{3}{5}$$

- (iii) Find the magnitude of the tension force in the rope in newtons (Leave your answer in terms of  $g$ ). 3

Horizontal forces:  $P = T\sin\theta$

Vertical forces:  $20g = T\cos\theta$

$$\tan\theta = \frac{P}{20g}$$

$$\frac{3}{4} = \frac{P}{20g}$$

$$P = 15g$$

$$15g = T\sin\theta$$

$$15g = \frac{3T}{5}$$

$$T = 25g \text{ newtons}$$

- (b) Prove that, if
- $a^2 - 2a + 2025$
- is even, then
- $a$
- is odd.

3

By contrapositive, or this is quicker:

If  $a^2 - 2a + 2025$  is even, then $a^2 - 2a$  is odd. $a(a - 2)$  is odd. $\therefore$  both  $a$  and  $a - 2$  are odd. $a$  is odd.

- (c) Use integration by parts to find an expression for

3

$$\int e^x \sin x \, dx.$$

$$\begin{aligned} u &= e^x, v' = \sin x \\ u' &= e^x, v = -\cos x \end{aligned}$$

$$I = uv - \int u'v$$

$$= -e^x \cos x - \int -e^x \cos x \, dx$$

$$= -e^x \cos x + \int e^x \cos x \, dx$$

$$= -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

$$2I = e^x(\sin x - \cos x)$$

$$I = \frac{e^x}{2}(\sin x - \cos x) + C$$

$$\begin{aligned} u &= e^x, v' = \cos x \\ u' &= e^x, v = \sin x \end{aligned}$$

- (d) Use the
- $t$
- formulas to find an expression for

4

$$\int \frac{1}{5 + 4 \cos x + 3 \sin x} \, dx.$$

$$\begin{aligned} dx &= \frac{2dt}{1+t^2} \\ \cos x &= \frac{1-t^2}{1+t^2} \\ \sin x &= \frac{2t}{1+t^2} \end{aligned}$$

$$= \int \frac{\frac{2dt}{1+t^2}}{5 + \frac{4-4t^2}{1+t^2} + \frac{6t}{1+t^2}} \times \frac{1+t^2}{1+t^2}$$

$$= \int \frac{2dt}{5 + 5t^2 + 4 - 4t^2 + 6t}$$

$$= \int \frac{2dt}{t^2 + 6t + 9}$$

$$= \int \frac{2dt}{(t+3)^2}$$

$$= -\frac{2}{t+3} + C$$

$$= -\frac{2}{\tan\left(\frac{x}{2}\right) + 3} + C$$

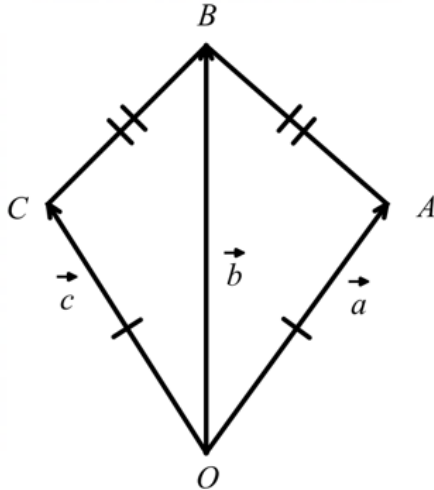
Type equation here.

**QUESTION 15** (16 marks) **Start a new writing booklet**

- (a)
- $OABC$
- is a kite, with
- $OB$
- as its line of symmetry.
- $|OA| = |OC|$
- .
- $|AB| = |BC|$
- .

Let  $\vec{a} = \vec{OA}$ ,  $\vec{b} = \vec{OB}$ ,  $\vec{c} = \vec{OC}$ .

- (i) Prove that
- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c}$
- .

**1**As the kite has line of symmetry  $OB$ , let  $\angle AOB = \angle COB = \theta$ .Also,  $|\vec{a}| = |\vec{c}|$  (properties of a kite).

$$\text{From } \triangle AOB, \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$\text{From } \triangle AOC, \cos \theta = \frac{\vec{c} \cdot \vec{b}}{|\vec{c}| |\vec{b}|} = \frac{\vec{c} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

As these are equal,  $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c}$ 

- (ii) Hence or otherwise, prove that the diagonals of a kite are perpendicular.

**3**From  $\triangle OCA$ ,  $\vec{CA} = \vec{a} - \vec{c}$ .

Dot product of diagonals:

$$\begin{aligned} & \vec{CA} \cdot \vec{OB} \\ &= (\vec{a} - \vec{c}) \cdot \vec{b} \\ &= \vec{a} \cdot \vec{b} - \vec{c} \cdot \vec{b} \\ &= \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{b} \text{ (from part (i))} \\ &= 0 \end{aligned}$$

Therefore, the diagonals are perpendicular.

(b) Use the principle of mathematical induction to prove that

3

$$(1 + x)^n > 1 + nx \text{ for } n > 1 \text{ and } x > 0$$

*Let  $n = 2$ :*

$$LHS = (1 + x)^2 = 1 + 2x + x^2$$

$$RHS = 1 + 2x$$

$$LHS > RHS \therefore \text{true for } n = 2$$

*Assume true for  $n = k$ , i.e.*

$$(1 + x)^k > 1 + kx \dots \dots (i)$$

*To prove true for  $n = k + 1$  i.e.*

$$(1 + x)^{k+1} > 1 + (k + 1)x \dots (ii)$$

*LHS of (ii)*

$$= (1 + x)^{k+1}$$

$$= (1 + x)(1 + x)^k$$

$$> (1 + x)(1 + kx) \text{ (using (i))}$$

$$= 1 + kx + x + kx^2$$

$$= 1 + (k + 1)x + kx^2$$

$$> 1 + (k + 1)x$$

Therefore true for  $n = k + 1$ , if true for  $n = k$ .

True for  $n = 2$ , hence by principle of mathematical induction, true for  $n = 3, 4, \dots \forall n$ .

- (c) Kajan decides to go bungee-jumping. This involves being tied to a bridge at point  $O$  by an elastic cable of length  $L$  metres, and then falling vertically from rest from this point.

After Kajan free falls  $L$  metres, he is slowed down by the cable, which exerts a force (in newtons) of  $mgk$  times the distance greater than  $L$  that he has fallen (where  $m$  is his mass in kg and  $k$  is a constant).

Let  $x$  metres be the distance Kajan has fallen, and let  $v$  m/s be his speed at  $x$ .

- (i) Show that  $\ddot{x} = \begin{cases} g & x \leq L \\ g - gk(x - L) & x > L \end{cases}$  2

When  $x \leq L$ , he is free falling, i.e.

$$\begin{aligned} m\ddot{x} &= mg \\ \ddot{x} &= g \end{aligned}$$

When  $x \geq L$ , after free falling,

$$\begin{aligned} m\ddot{x} &= mg - mgk(x - L) \\ \ddot{x} &= g - gk(x - L) \end{aligned}$$

- (ii) Show that  $v^2 = 2gL$  at the instant Kajan is at  $x = L$ . 1

$$\begin{aligned} \frac{d}{dx} \left( \frac{1}{2} v^2 \right) &= g \\ \frac{1}{2} v^2 &= gx + C \\ \text{at } x = 0, v = 0 &\rightarrow C = 0 \\ v^2 &= 2gx \\ \text{at } x = L, v^2 &= 2gL \end{aligned}$$

- (iii) Show that  $v^2 = 2gx - gk(x - L)^2$  for  $x > L$ . 1

$$\begin{aligned} \ddot{x} &= g - gk(x - L) \\ \frac{d}{dx} \left( \frac{1}{2} v^2 \right) &= g - gk(x - L) \\ \frac{1}{2} v^2 &= gx - \frac{gk(x - L)^2}{2} + C \\ \text{at } x = L, v = \sqrt{2gL} &\rightarrow C = 0 \\ v^2 &= 2gx - gk(x - L)^2 \end{aligned}$$

- (iv) Show that Kajan stops for the first time at  $x = L + \frac{1}{k} + \sqrt{\frac{2L}{k} + \left(\frac{1}{k}\right)^2}$ . 3

we want x when  $v = 0$

$$\begin{aligned}
 2gx - gk(x - L)^2 &= 0 \\
 2gx - gk(x^2 - 2xL + L^2) &= 0 \\
 2gx - gkx^2 + 2gkLx - gkL^2 &= 0 \\
 kx^2 - 2(1 + kL)x + kL^2 &= 0 \\
 x^2 - \frac{2(1 + kL)}{k}x + L^2 &= 0 \\
 x^2 - \frac{2(1 + kL)}{k}x + \frac{(1 + kL)^2}{k^2} &= \frac{(1 + kL)^2}{k^2} - L^2 \\
 \left(x - \frac{(1 + kL)}{k}\right)^2 &= \frac{(1 + kL)^2 - k^2L^2}{k^2} \\
 \left(x - \frac{(1 + kL)}{k}\right)^2 &= \frac{1 + 2kL}{k^2} \\
 x - \frac{(1 + kL)}{k} &= \frac{\sqrt{1 + 2kL}}{k} \\
 x &= \frac{1 + kL}{k} + \frac{\sqrt{1 + 2kL}}{k} \\
 x &= \frac{1}{k} + L + \sqrt{\frac{1 + 2kl}{k^2}} \\
 x &= \frac{1}{k} + L + \sqrt{\frac{2L}{k} + \frac{1}{k^2}}
 \end{aligned}$$

- (v) Given that  $\frac{1}{k} = \frac{L}{4}$ , show that  $O$  must be at least  $2L$  metres above any 2

obstruction on Kajan's path.

$$\begin{aligned}
 \frac{1}{k} &= \frac{L}{4} \\
 \therefore x &= \frac{L}{4} + L + \sqrt{\frac{L}{4}(2L) + \frac{L^2}{16}} \\
 x &= \frac{L}{4} + L + \sqrt{\frac{L^2}{2} + \frac{L^2}{16}} \\
 x &= \frac{L}{4} + L + \sqrt{\frac{9L^2}{16}} \\
 x &= \frac{L}{4} + L + \frac{3L}{4} \\
 x &= 2L
 \end{aligned}$$

**QUESTION 16 (15 marks) Start a new writing booklet**

(a) Evaluate

**4**

$$\begin{aligned}
 & \int_1^2 \frac{x+1}{\sqrt{3x-2-x^2}} dx \\
 &= -\frac{1}{2} \int_1^2 \frac{-2x+3}{\sqrt{3x-2-x^2}} dx + \frac{5}{2} \int_1^2 \frac{1}{\sqrt{3x-2-x^2}} dx \\
 &= -\frac{1}{2} \int_1^2 \frac{-2x+3}{\sqrt{3x-2-x^2}} dx + \frac{5}{2} \int_1^2 \frac{1}{\sqrt{\frac{1}{4} - \left(x^2 - \frac{3}{2}\right)^2}} dx \\
 &= \left[ -\frac{1}{2} \times 2\sqrt{3x-2-x^2} \right]_1^2 + \left[ \frac{5}{2} \sin^{-1}(2x-3) \right]_1^2 \\
 &= 0 + \frac{5}{2} (\sin^{-1} 1 - \sin^{-1}(-1)) \\
 &= \frac{5}{2} \left( \frac{\pi}{2} - \left( -\frac{\pi}{2} \right) \right) \\
 &= \frac{5\pi}{2}
 \end{aligned}$$

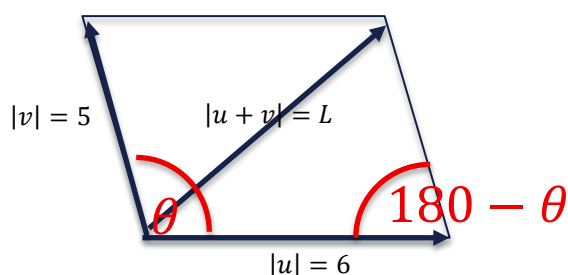
(b) Find  $|u + v|$  if  $|u| = 6$ ,  $|v| = 5$  and  $u \cdot v = -4$ .

3

$$\begin{aligned}
 (u + v) \cdot (u + v) &= |u + v|^2 = |u|^2 + 2u \cdot v + |v|^2 \\
 |u + v|^2 &= |u|^2 + 2u \cdot v + |v|^2 \\
 |u + v|^2 &= 36 - 8 + 25 \\
 |u + v|^2 &= 53 \\
 |u + v| &= \sqrt{53}
 \end{aligned}$$

Alternatively, this can be modelled by a parallelogram with adjacent sides  $|u| = 6$ ,  $|v| = 5$ , and long diagonal.

$|u + v|$ .



Let the included angle  $\theta$  be given by

$$\begin{aligned}
 \cos \theta &= \frac{u \cdot v}{|u||v|} \\
 &= \frac{-4}{30}
 \end{aligned}$$

Hence the other angle is  $180 - \theta$  and its cosine is  $+\frac{2}{15}$

By the cosine rule,

$$\begin{aligned}
 L^2 &= |u|^2 + |v|^2 - 2|u||v|\cos(180 - \theta) \\
 &= 36 + 25 - 2 \times 6 \times 5 \times \frac{4}{30} \\
 &= 53 \\
 |u + v| &= \sqrt{53}
 \end{aligned}$$

(c) Prove that

2

$$\frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} = \frac{(n+1)!}{r!(n-r+1)!}$$

*LHS has an LCD of  $r!(n-r+1)!$*

$$\begin{aligned} LHS &= \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} \\ &= \frac{n!(n-r+1) + n!r}{r!(n-r+1)!} \\ &= \frac{n!(n+1)}{r!(n-r+1)!} \\ &= \frac{(n+1)!}{r!(n-r+1)!} \\ &= RHS \text{ as required} \end{aligned}$$

- (d) It is known that the arithmetic mean of three numbers is greater than the geometric mean of those numbers, i.e.

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}.$$

- (i) Given  $a+b+c = t$  and  $a, b, c > 0$ , prove that

3

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{t}.$$

$$\frac{a+b+c}{3} \geq (abc)^{\frac{1}{3}} \quad (\text{by AM} \geq \text{GM})$$

$$a+b+c \geq 3(abc)^{\frac{1}{3}}$$

$$t \geq 3(abc)^{\frac{1}{3}}$$

$$\frac{t}{3} \geq (abc)^{\frac{1}{3}}$$

$$\frac{3}{t} \leq \frac{1}{(abc)^{\frac{1}{3}}}$$

$$\frac{1}{(abc)^{\frac{1}{3}}} \geq \frac{3}{t}$$

Now,

$$\frac{\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)}{3} \geq \left\{\frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c}\right\}^{\frac{1}{3}} \quad (\text{by AM} \geq \text{GM})$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3 \left\{\frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c}\right\}^{\frac{1}{3}}$$

$$\geq 3 \left\{\frac{1}{abc}\right\}^{\frac{1}{3}}$$

$$\geq 3 \frac{1}{(abc)^{\frac{1}{3}}}$$

$$\geq 3 \cdot \frac{3}{t}$$

$$\geq \frac{9}{t}$$

(ii) Hence, given  $a, b, c > 0$  and  $a + b + c = 1$ , prove that**3**

$$\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) \geq 8.$$

$$\begin{aligned} \left\{\frac{1}{a} - 1\right\} \cdot \left\{\frac{1}{b} - 1\right\} \cdot \left\{\frac{1}{c} - 1\right\} &= \frac{(1-a)(1-b)(1-c)}{abc} \\ &= \frac{1 - (a+b+c) + (ab+ac+bc) - (abc)}{abc} \end{aligned}$$

Since  $a + b + c = 1$ 

$$\begin{aligned} &= \frac{ab+ac+bc-abc}{abc} \\ &= \frac{1}{a} + \frac{1}{b} + \frac{1}{c} - 1 \end{aligned}$$

But  $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$  from (i)

$$\therefore \left\{\frac{1}{a} - 1\right\} \cdot \left\{\frac{1}{b} - 1\right\} \cdot \left\{\frac{1}{c} - 1\right\} \geq 8.$$

**END OF PAPER**

Student Name: \_\_\_\_\_

**2025****Year 12 Mathematics Extension 2**

Trial HSC Examination

**MULTIPLE-CHOICE ANSWER SHEET**

For multiple choice questions, choose the best answer A, B, C or D and fill in the correct circle.

1.    ☐ A   ☐ B   ☐ C   ☐ D
2.    ☐ A   ☐ B   ☐ C   ☐ D
3.    ☐ A   ☐ B   ☐ C   ☐ D
4.    ☐ A   ☐ B   ☐ C   ☐ D
5.    ☐ A   ☐ B   ☐ C   ☐ D
6.    ☐ A   ☐ B   ☐ C   ☐ D
7.    ☐ A   ☐ B   ☐ C   ☐ D
8.    ☐ A   ☐ B   ☐ C   ☐ D
9.    ☐ A   ☐ B   ☐ C   ☐ D
10.   ☐ A   ☐ B   ☐ C   ☐ D